

COMPUTER SCIENCE TRIPOS Part IB – 2024 – Paper 6

7 Logic and Proof (mj201)

Modal Logic

A client wants a dispensing machine that sells two products, A and B . The following are some requirements:

1. In the *default state* a screen displays the two unselected products and it assumes no payment has been made.
2. Every time the machine is in the default state the user may select one and only one product.
3. It is always true that, given a selection, the user may pay for the selected product.
4. The selection can be changed provided that the user has not paid.
5. It is always possible to return to the default state.

Once a product has been selected and paid for, the user gets the product.

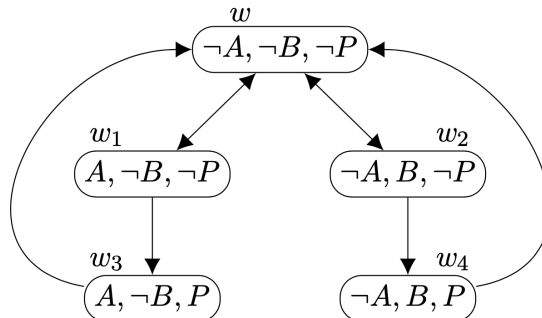
- (a) Formalise statements 1 to 5 in S4 modal logic. Use letter A to represent the proposition that A is selected, B to represent that B is selected, D to represent the default state and P that a payment has been made. [6 marks]

Answer: Let A and B denote the product selection, let D be the default state, and let P denote that a payment has been made. Thus, the default state is characterised by $\neg A$, $\neg B$ and $\neg P$.

1. $\neg A \wedge \neg B \wedge \neg P$ (alternatively, $\Box[D \leftrightarrow (\neg A \wedge \neg B \wedge \neg P)]$ is acceptable, also $D \rightarrow$ here would be accepted),
 2. $\Box[(\neg A \wedge \neg B \wedge \neg P) \rightarrow \Diamond(A \wedge \neg B)]$, $\Box[(\neg A \wedge \neg B \wedge \neg P) \rightarrow \Diamond(\neg A \wedge B)]$, (alternatively, $\Box[D \rightarrow \Diamond(A \wedge \neg B)]$ and $\Box[D \rightarrow \Diamond(\neg A \wedge B)]$ are acceptable, or the conjunction of these)
 3. $\Box(A \rightarrow \Diamond(A \wedge P))$, $\Box(B \rightarrow \Diamond(B \wedge P))$,
 4. $\Box(\neg P \rightarrow \Diamond A \wedge \Diamond B)$,
 5. $\Box\Diamond(\neg A \wedge \neg B \wedge \neg P)$ (alternatively, $\Box\Diamond D$ is acceptable).
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- (b) Model the default state as a world in a modal frame. Exhibit an interpretation that satisfies statements 1 to 5. [7 marks]

Answer:



- (c) Given your formalisation of statements 1 to 5, how would you formalise and prove (using the sequent calculus for S4) that, starting from the default state, it is possible for the user to get one of every product? Your answer can be schematic. A full formal proof is not necessary, but you must be clear about the strategy used. [*Hint*: Start by using the default state and statement 2 to prove that it is possible to select A .] [7 marks]

Answer: We formalise this statement as follows: $\Diamond(A \wedge P \wedge \Diamond(B \wedge P))$. It means that there exists a future where we have selected and paid for A and in that future it is still possible to select and pay for B .

The main strategy of the proof is to use 1. and 2. to jump to a future where A has been selected, then use 3. to jump to a future where it has been paid for. Then we use 5. to go back to the default state, from which we can again use 1. and 2. to jump to a future where B as been selected, then use 5. to jump to a future where it has been paid for.

For simplicity let Γ be the set of statements 2., 3., and 5. (statement 4. is not needed). It is important to note that they are all wrapped by the $\Box$ operator, which means that they are unaffected by any use of the rules $(\Box r)$ or $(\Diamond l)$.

First, it is easy to prove $\Diamond(A \wedge \neg B)$ from statements 1. and 2. Thus we omit that proof and use $\Diamond(A \wedge \neg B)$ as an assumption:

$$\begin{array}{c}
 \vdots \\
 \hline
 \neg A \wedge \neg B \wedge \neg P, \Gamma \Rightarrow \Diamond(B \wedge P) \\
 \hline
 \Diamond(\neg A \wedge \neg B \wedge \neg P), \Gamma \Rightarrow \Diamond(B \wedge P) \quad (\Diamond l) \\
 \hline
 \Box \Diamond(\neg A \wedge \neg B \wedge \neg P), \Gamma \Rightarrow \Diamond(B \wedge P) \quad (\Box l) \\
 \hline
 \frac{A \wedge P, \Gamma \Rightarrow A \wedge P \quad (\text{basic})}{A \wedge P, \Gamma \Rightarrow \Diamond(B \wedge P)} \quad \text{assm 5.} \\
 \hline
 \frac{A \wedge P, \Gamma \Rightarrow A \wedge P \wedge \Diamond(B \wedge P)}{A \wedge P, \Gamma \Rightarrow \Diamond(A \wedge P \wedge \Diamond(B \wedge P))} \quad (\wedge r) \\
 \hline
 \frac{A \wedge P, \Gamma \Rightarrow \Diamond(A \wedge P \wedge \Diamond(B \wedge P))}{A \wedge P, \Gamma \Rightarrow \Diamond(A \wedge P \wedge \Diamond(B \wedge P))} \quad (\Diamond r) \\
 \hline
 \frac{A, \dots \Rightarrow A, \dots \quad (\text{basic})}{\Diamond(A \wedge P), \Gamma \Rightarrow \Diamond(A \wedge P \wedge \Diamond(B \wedge P))} \quad (\Diamond l) \\
 \hline
 \frac{A, \neg B, A \rightarrow \Diamond(A \wedge P), \Gamma \Rightarrow \Diamond(A \wedge P \wedge \Diamond(B \wedge P))}{A \wedge \neg B, \Box(A \rightarrow \Diamond(A \wedge P)), \Gamma \Rightarrow \Diamond(A \wedge P \wedge \Diamond(B \wedge P))} \quad (\rightarrow l) \\
 \hline
 \frac{A \wedge \neg B, \Box(A \rightarrow \Diamond(A \wedge P)), \Gamma \Rightarrow \Diamond(A \wedge P \wedge \Diamond(B \wedge P))}{\Diamond(A \wedge \neg B), \Box(A \rightarrow \Diamond(A \wedge P)), \Gamma \Rightarrow \Diamond(A \wedge P \wedge \Diamond(B \wedge P))} \quad (\Box l, \wedge l) \\
 \hline
 \frac{\Diamond(A \wedge \neg B), \Box(A \rightarrow \Diamond(A \wedge P)), \Gamma \Rightarrow \Diamond(A \wedge P \wedge \Diamond(B \wedge P))}{\Diamond(A \wedge \neg B), \Box(A \rightarrow \Diamond(A \wedge P)), \Gamma \Rightarrow \Diamond(A \wedge P \wedge \Diamond(B \wedge P))} \quad (\Diamond l)
 \end{array}$$

The rest of the proof, indicated by vertical ellipsis, is similar to the proof already shown. The assumptions are the same as in the default state, so we can obtain $\Diamond(\neg A \wedge B)$ instead of $\Diamond(A \wedge \neg B)$ and use it to prove $\Diamond(B \wedge P)$. If the default state is modelled by D , the axioms $\Box[D \leftrightarrow (\neg A \wedge \neg B \wedge \neg P)]$ and $\Box \Diamond D$ allow us to jump to the state where D holds in order to use $\neg A \wedge \neg B \wedge \neg P$.

This model answer is very extensive. This high level of detail is not required for full marks, however the proof strategy must be made clear, and the main steps must be shown clearly.